

Chem 210D: Modern Applications of Inorganic Chemistry

Unit 1: Nuclear and Quantum Chemistry

Fall Semester 2026

Recommended Reading in Tro

Chapter 20 (Nuclear Chemistry): Sections 20.2, 20.3, 20.4, 20.6, 20.7, and 20.8

Chapter 2 (Quantum chemistry): Sections 2.3 - 2.6, Cox Handout on Particle-in-a-Box

Chapter 3 (Periodic Trends): Sections 3.3 - 3.8, Cox Handout on Relativistic Effects

Graded Mastering Chemistry Homework

All homework is due on Tuesday at 11:59 pm. Each assignment is worth 12 points.

September 8: Nuclear Chemistry

September 15: Quantum Chemistry

September 22: Periodic Trends

Extra Practice Problems

These problems with solutions are available on Canvas under the Modules tab. Look for "Extra Practice by Topic."

There are two sets of practice problems: 1-CHEM 210 Nuclear Chemistry Extra Practice & 2-CHEM 210 Quantum Chemistry Extra Practice.

Additionally, we will have problem-solving office hours on August 31, September 3, September 14, and September 17th covering concepts related to nuclear chemistry, quantum chemistry, and periodic trends.

Qualitative and Quantitative Concepts for Mastery

- 1.) Explain why nuclear decay occurs. Identify the pathway one would expect based on the band of stability curve. What pathway is favored for nuclei that lie above and below the band of stability?
- 2.) What is a mass defect? How is the mass defect used to calculate the energy released during binding and nuclear reactions? Why are nuclear reactions typically exothermic?
- 3.) Use half-life and activity for calculations related to nuclear decay.
- 4.) What are the forces that act on electrons based on the Bohr model? What is the relationship between kinetic energy and potential energy based on nuclear charge?
- 5.) What is duality? How can we use the de Broglie wavelength to determine whether duality is probable?
- 6.) Describe the difference between classical and quantum physics.
- 7.) From the particle-in-a-Box model explain the relationship between energy and principal quantum

number (n), energy and space the electron occupies (L), energy and nodes, and nodes and principal quantum number (n).

8.) Identify the two types of nodes. Why does a 1p orbital not exist?

9.) Use a Radial Distribution Curve to explain why electrons in s-orbitals often "feel" greater nuclear charge.

10.) Calculate effective nuclear charge using Slater's Rules.

11.) Predict trends in ionization energy using effective nuclear charge and the principal quantum number (n).

12.) Explain the concept of lanthanide contraction to account for the fact that 4d and 5d metals are similar in size.

13.) Define electron affinity.

14.) Explain why Gold, Mercury, Thallium, and Lead have unique properties. Specifically, account for the electronegativity of gold, why mercury acts like a noble gas, and the inert pair effect observed with thallium and lead.

Miscellaneous Resources Linked on Canvas

Phet Simulations: Coulomb's Law & Isotopes and Atomic Mass

Bohr Model Derivations (derivations will not be tested)

The Orbitron (to Visualize Orbitals) (available under the pages tab)

The Orbitron is a great resource to help visualize radial and angular nodes.